



Computer Systems Engineering Technology Department
CST 407 – Python for AI

Exam

In this exam you will be investigating the classification of **seventeen distinct cinnamon** samples utilizing Quartz Crystal Microbalance (QCM) sensors modified with **ten different** metal oxide nanoparticle dispersions and polymers. Sensor outputs and predictions are given in “cinnamoncut-org.csv” file. File has 1938 data points.

1. Get familiar with the data set: Read the file in python, determine number of classes, determine how many data points are available from each class etc.
2. Now split the data into two parts: training.csv, test.csv
While splitting the data make sure that each class is represented in both files with enough data points. The optimum number of data points on training.csv and test.csv should be determined by you.
3. Make the necessary data and label modifications so that you can feed it to your NN structure.
4. Construct your first NN structure. Only use dense layers. Determine the number of layers, the activation functions for layers, loss criteria, optimizer and other parameters needed.
5. Run your first NN structure and plot the confusion matrix. Comment on first results.
6. Now use keras tuner to determine optimum number of dense layers and optimum number of neurons on each layer.
7. Choose the best NN structure determined by the keras tuner and test it with test data.
8. Plot the confusion matrix
9. Compare the result with your first NN structure's result.
10. Submit both your jupyter notebook, report, test.csv and train.csv (zip and rar in a single folder)